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Brillouin Scattering in Liquids

G. BENEDEK AND T. GREYTAK

Abstract—The development of monochromatic, intense, and highly directional laser light sources has made it possible to measure accurately the velocity, the frequency, and the lifetimes of thermally excited hypersonic sound waves. The technique employed is to study the spectrum of the light scattered from the thermally generated sound waves in the medium.

A theory is presented which demonstrates explicitly the information contained in the intensity, the spectral positions, and spectral widths of the scattered light. An experimental system is also described to measure the spectrum. Preliminary data is presented on the velocity and lifetimes of ~ 6 Gc/s sound waves in water and toluene.

INTRODUCTION

THE EXPERIMENTAL STUDY of the propagation of sound typically requires an acoustic transducer. This device has the dual purpose of generating a sound wave and then detecting this wave after it has traveled in the medium. Although it may seem that the transducer is indispensable, it is worth remembering that in exciting a sound wave, the transducer simply adds a single-frequency high-intensity wave to the assemblage of thermally excited sound waves which are already present in the medium. In a sense we can look upon the heat content of a medium as a "thermal exciter." This "thermal exciter" is remarkable in that it produces not a single sound wave, but rather a myriad of sound waves whose wavelengths λ range from $\lambda \sim$ the size of the container to $\lambda \sim$ interatomic distances. In the effort to study sound of ever

higher frequency, we should keep in mind that by thermal excitation nature provides us, gratis, with sound waves of frequency ranging from a few hundred kilocycles per second to about 1000 gigacycles per second.

What is needed, of course, is some means of detecting these thermal sound waves, and determining their properties: Brillouin's work [1], [2] showed that light waves can be used for this purpose. If a light wave of wavelength λ_0 is incident on a medium whose index of refraction is n , then the light scattered through an angle θ away from the incident beam is the result of a "reflection" off a sound wave whose wavelength λ_s is

$$\lambda_s = \lambda_0 / (2n \sin \frac{1}{2}\theta). \quad (1)$$

From this formula we see that the shortest sound wavelength responsible for light scattering is equal to one half the wavelength of light in the medium. The spectrum of the scattered light consists of a doublet split symmetrically around the incident frequency by an amount $\pm \Omega(\theta)$. The splitting of these "Brillouin doublets" is equal to the frequency of the scattering sound wave. If V_s is the sound wave velocity, $\Omega(\theta)$ is given by

$$(\Omega(\theta)/2\pi) = V_s/\lambda_s = (2V_s n \sin(\frac{1}{2}\theta))/\lambda_0. \quad (2)$$

Thus, the scattering angle θ determines the wavelength of the scattering sound wave, and the splitting of the doublet gives the frequency of that sound wave. Furthermore, the spectral width of the Brillouin doublets gives the lifetime of the sound wave.

It is possible then to do without the transducer, for heat to generate the sound waves and light to detect them. This technique, known as Brillouin scattering, has in fact, been used extensively, particularly by

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Russian [3] and Indian workers, for the study of thermally excited sound waves both in liquids and solids. Until recently, however, the full promise of this technique was not fulfilled because of the lack of monochromaticity and the relatively low intensity of conventional light sources. The breadth of the spectrum of the incident light made it difficult to resolve the Brillouin doublets and almost impossible to determine their natural line width. The advent of the laser has radically altered this state of affairs. These new light sources are extremely monochromatic, intense, and unidirectional. Recent experiments [4], [5] have shown them to be very valuable in Brillouin scattering. We now stand at a point where we can look forward to using these laser light sources in conjunction with high resolution spectroscopy to determine the velocity and lifetime of sound waves whose frequency is as high as 10 Gc/s in liquids and 100 Gc/s in solids.

The theory of Brillouin scattering in liquids is outlined here. In particular, the intensity and the detailed form of the spectrum of the scattered light are calculated and measurements of the sound velocity and lifetime for ~ 6 Gc/s sound waves in water and toluene are presented.

THEORY

It is helpful at the outset to examine the rudiments of the theory of the scattering of light from an elementary point of view. As a light wave passes through a medium it generates an oscillating dipole moment at each point which radiates or scatters electromagnetic energy away in every direction. The vector sum of the fields scattered from each point in the medium is zero in any direction other than the forward direction if the medium is perfectly homogeneous. If the medium, on the other hand, contains a periodic modulation in its scattering power a "Bragg reflection" off this modulation can occur. Bragg's law shows that this constructive interference takes place if the wavelength of the modulation (λ_s) is related to the wavelength of light in the medium (λ_0/n) by (1), where θ is the angle between the wave vectors of the incident and scattered light waves $\mathbf{k}$ and $\mathbf{k}'$. In addition to the scalar condition in (1), Bragg's law requires that the wavefronts of the scattering medium must be planes that reflect $\mathbf{k}$ into $\mathbf{k}'$ as shown in Fig. 1. We may write Bragg's law vectorially in terms of $\mathbf{k}$, $\mathbf{k}'$ and $\mathbf{K}$, the wave vector of the scattering wave, as

$$\mathbf{k}' - \mathbf{k} = \mathbf{K}. \quad (3)$$

This equation, which has the form of a momentum conservation condition between the light waves and the matter wave, is identical to Bragg's law, provided that the wavelength of the scattered light is nearly equal to that of the incident light. Under this condition

$$|\mathbf{k}' - \mathbf{k}| = 2|\mathbf{k}| \sin \frac{1}{2}\theta = 2(2\pi n/\lambda_0) \sin \frac{1}{2}\theta. \quad (4)$$

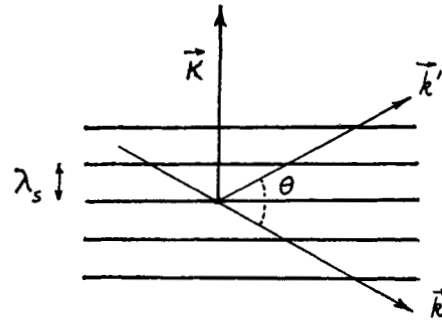


Fig. 1. Representation of the scattering as a Bragg reflection off the modulation in the index of refraction produced by the thermally excited sound waves.

Since $|\mathbf{K}| = (2\pi/\lambda_s)$, it follows that (3) is Bragg's law. By scattering through larger and larger angles one observes the effect of shorter and shorter wavelength sound waves, the shortest being $\lambda_s = \lambda_0/2n$.

The motion of the sound wave through the medium has a profound effect on the spectrum of the light reflected off the sound waves. We can find the effect of this motion by imagining that the wavefronts shown in Fig. 1 are moving in the directions $\pm \mathbf{K}$. The light wave is reflected off the *moving* sound wave and thereby suffers a Doppler shift proportional to twice the phase velocity of the sound wave. More accurately, the Doppler shift is determined by the component of the sound velocity in the direction of propagation of the light wave, i.e., by $V_s \cos(90 - \theta/2)$. Since (c/n) is the velocity of light in the medium, the Doppler shift in the light scattered through an angle θ is

$$(\Delta\omega/\omega_0) = \pm 2(V_s/c)n \sin \frac{1}{2}\theta \quad (5)$$

where ω_0 is the frequency of the incident light wave. This formula was first found by L. Brillouin [1], [2]. It shows that the spectrum of the scattered light contains a doublet due to the Doppler shift of the advancing and receding sound waves which reflected the light wave. The splitting of the doublet gives, through (5), the velocity of the reflecting sound wave. Alternatively, the splitting of the Brillouin doublets is just equal to the frequency of the sound wave responsible for the scattering, as can be seen by using (1) to eliminate $\sin \frac{1}{2}\theta$ from (5). This frequency ranges from ~ 0 for forward scattering $K=0$, to ~ 10 Gc/s for backward scattering $K \sim (2\pi/(\lambda_0/2n))$. Thus the scattering angle fixes the wavelength of the sound wave while the splitting of the doublet gives the frequency Ω of this sound wave. By determining Ω for various values of K , we may determine the dispersion relation for the sound waves.

The elementary analysis just discussed has the limitations that it cannot give the intensity of the scattering, nor does it provide the width or the shape of the spectrum of the scattered light. For this information we now turn to a rigorous classical theory of the scattering. Let the electric field ($E_{inc}(r, t)$) of the incident light wave in the medium have the plane wave form

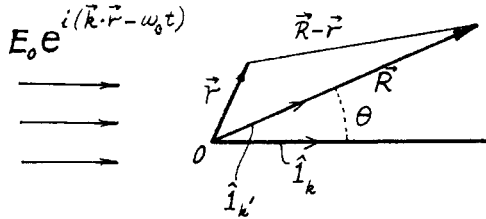


Fig. 2. Diagram of the scattering geometry where r is the source point, R the field point, $\hat{i}_k$ and $\hat{i}_{k'}$ are, respectively, unit vectors in the direction of the incident and scattered light beams.

$$E_{\text{inc}}(r, t) = E_0 \exp [i(k \cdot r - \omega_0 t)] \quad (6)$$

as indicated in Fig. 2. The light wave produces an oscillating polarization $P(r, t)$ at each point (r) in the medium. The oscillating polarization in the volume $|dr|$ radiates an electric field whose value at the field point R is given by

$$dE_{\text{scatt}}(R, t) = \frac{\ddot{P}\left(r, t - \frac{|R - r|}{c}\right) \sin \phi |dr|}{c^2 |R - r|} \quad (7)$$

where $\sin \phi$ is the angle between the polarization vector of the incident wave and the propagation vector of the scattered wave. $P(r, t)$ is related to the incident field by the polarizability α or the dielectric constant ϵ . In the presence of the thermally excited sound waves both these quantities must be regarded as functions of position and time; hence

$$P(r, t') = \alpha(r, t') E_{\text{inc}}(r, t') = \frac{[\epsilon(r, t') - 1]}{4\pi} E_{\text{inc}}(r, t'). \quad (8)$$

We may now calculate the second derivative of the polarization from (8), neglecting the time dependence of α in comparison with the very rapid oscillations in the incident field. If, furthermore, we take $R \gg r$ and neglect the small change in the wavelength in the scattered light, then the electric field at R scattered by the entire illuminated volume is simply

$$E_{\text{scatt}}(R, t) \cong E_0 \left(\frac{\omega_0}{c}\right)^2 \frac{\exp [i(k' \cdot R - \omega_0 t)] \sin \phi}{R} \cdot \int_V \alpha(r, t) \exp [i(k - k') \cdot r] |dr| \quad (9)$$

where

$$k' = \hat{i}_{k'} [2\pi/(\lambda_0/n)]$$

and where V is the illuminated volume. The scattered field is a spherically spreading wave whose amplitude is determined by the interference integral

$$I = \int_V \alpha(r, t) \exp [-iK \cdot r] |dr| \quad (10)$$

where the scattering vector K is defined by

$$K = k' - k. \quad (11)$$

This interference integral represents the superposition of phases of waves scattered from each point in the medium. If we break the polarizability into its average value plus the fluctuations around the average, i.e.,

$$\alpha(r, t) = \langle \alpha \rangle + \delta \alpha(r, t) \quad (12)$$

we observe on substituting (12) into (10) that the uniform part of the polarizability produces no scattering, except in the forward direction $K = 0$. The scattering in directions other than the forward direction originates entirely in the fluctuations in the polarizability $\delta \alpha(r, t)$. We can find the fluctuation which is responsible for scattering in the direction k' by making a Fourier decomposition of the fluctuations in α , viz,

$$\delta \alpha(r, t) = \frac{1}{(2\pi)^{3/2}} \int |dq| \exp [iq \cdot r] \delta \alpha(q, t). \quad (13)$$

Substituting this into (10) we see that

$$I = \frac{1}{(2\pi)^{3/2}} \int |dq| \delta \alpha(q, t) \int_V \exp [i(q - K) \cdot r] |dr|. \quad (14)$$

But the last integral in (14) is zero unless $q = K$, in fact

$$\int \exp [i(q - K) \cdot r] |dr| = (2\pi)^3 \delta(q - K). \quad (15)$$

Thus the scattering in the direction k' is produced by a fluctuation whose wave vector q is equal to the scattering vector K . This is the Bragg reflection condition, or the conservation of momentum condition discussed previously. We are now in a position to write the scattered field in terms of the Fourier amplitude for the fluctuation in α or ϵ . Using (15), (14), (9), and $\delta \alpha = \delta \epsilon / 4\pi$, we find the fundamental formula for the scattered field:

$$E_{\text{scatt}}(R, t) = E_0 \left(\frac{\omega_0}{c}\right)^2 \frac{\sin \phi}{4\pi} \frac{\exp [i(k \cdot R - \omega_0 t)]}{R} (2\pi)^{3/2} \delta \epsilon(K, t). \quad (16)$$

To find the intensity and the spectrum of the scattered light it remains for us to calculate the amplitude and the time dependence of $\delta \epsilon(K, t)$, the Fourier transform of the fluctuation in the dielectric constant $\delta \epsilon(r, t)$. These quantities are related by the inverse of (13), i.e.,

$$\delta \epsilon(K, t) = \frac{1}{(2\pi)^{3/2}} \int |dr| \exp [-iK \cdot r] \delta \epsilon(r, t). \quad (17)$$

If we regard ϵ as a function of the local density ρ and the local temperature T we can see that as the sound wave courses through the medium it produces fluctuations in ρ and T and hence fluctuations in ϵ . Thus we write

$$\delta \epsilon(r, t) = \left(\frac{\partial \epsilon}{\partial \rho}\right)_T \delta \rho(r, t) + \left(\frac{\partial \epsilon}{\partial T}\right)_\rho \delta T(r, t). \quad (18)$$

The fluctuations $\delta\rho$ and δT are related to each other when they are produced by a sound wave, because the entropy does not change. It is convenient to define a local entropy density $S^* = (S/V_0)$ where V_0 is a small fixed volume surrounding the point r . As the sound wave passes, S^* , which we take as a function of ρ and T , remains constant, i.e.,

$$\delta S^* = \left(\frac{\partial S^*}{\partial \rho}\right)_T \delta\rho(r, t) + \left(\frac{\partial S^*}{\partial T}\right)_\rho \delta T(r, t) = 0 \quad (19)$$

so that

$$\delta T(r, t) = \frac{-\left(\frac{\partial S^*}{\partial \rho}\right)_T}{\left(\frac{\partial S^*}{\partial T}\right)_\rho} \delta\rho(r, t). \quad (20)$$

From the definition of the specific heat at constant volume and the Maxwell relation

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V \quad (21)$$

it follows at once that (20) can be written as

$$\delta T(r, t) = \left(\frac{a}{\beta_T}\right) \frac{T}{\rho_0 c_V^*} \delta\rho(r, t) \quad (22)$$

where $a = (1/V)(\partial V/\partial T)_T$ is the thermal expansion coefficient, $\beta_T = -(1/V)(\partial V/\partial P)_T$ is the isothermal compressibility, c_V^* is the specific heat per unit volume, and ρ_0 is the mean density. Thus the fluctuations in the dielectric constant which come from the sound wave can be expressed entirely in terms of the density fluctuation in the wave by combining (22) and (18), thus giving

$$\delta\epsilon(r, t) = \left[\left(\frac{\partial\epsilon}{\partial\rho}\right)_T + \left(\frac{\partial\epsilon}{\partial T}\right)_\rho \left(\frac{a}{\beta_T}\right) \frac{T}{\rho_0 c_V^*}\right] \delta\rho(r, t). \quad (23)$$

$\delta\rho(r, t)$ is connected to the displacement $U(r, t)$ in the sound wave by the continuity equation

$$\delta\rho(r, t) = \rho_0 \nabla \cdot U(r, t). \quad (24)$$

A relation between $\delta\epsilon(K, t)$ and the sound wave amplitudes can now be obtained directly. Equations (17) and (23) show that $\delta\epsilon(K, t)$ is proportional to $\delta\rho(K, t)$. This latter quantity according to (24) is given by

$$\delta\rho(K, t) = \frac{1}{(2\pi)^{3/2}} \int |d\mathbf{r}| \exp[-i\mathbf{K} \cdot \mathbf{r}] \rho_0 \nabla \cdot U(\mathbf{r}, t). \quad (25)$$

This integral can be considerably simplified by decomposing the sound wave displacement $U(\mathbf{r}, t)$ into its Fourier components $U(\mathbf{q}, t)$, viz,

$$U(\mathbf{r}, t) = \frac{1}{(2\pi)^{3/2}} \int |d\mathbf{q}| \exp[i\mathbf{q} \cdot \mathbf{r}] U(\mathbf{q}, t). \quad (26)$$

On substituting this into (25), changing the order of integration and using the δ function property in (15),

we obtain

$$\delta\rho(K, t) = i\rho_0 \mathbf{K} \cdot U(\mathbf{K}, t). \quad (27)$$

Thus, from (17), (23), and (27) we have the desired relation

$$\delta\epsilon(K, t) = \left[\left(\frac{\partial\epsilon}{\partial\rho}\right)_T + \left(\frac{\partial\epsilon}{\partial T}\right)_\rho \left(\frac{a}{\beta_T}\right) \frac{T}{\rho_0 c_V^*}\right] i\rho_0 \mathbf{K} \cdot U(\mathbf{K}, t). \quad (28)$$

Since the sound waves in a liquid are longitudinal, $\mathbf{K}$ and $\mathbf{U}$ are parallel and $\mathbf{K} \cdot U(\mathbf{K}, t) = K U(\mathbf{K}, t)$.

We see from (28) and (16) that to find the scattered electric field we need to know the magnitude and time dependence of the Fourier amplitudes of the thermally excited sound waves. These waves grow spontaneously, oscillate at the sound frequency $\Omega(\mathbf{K})$, and decay away at a characteristic rate $\Gamma(\mathbf{K})$ only to be replaced by another wave with a different starting phase and a similar temporal history. The Fourier amplitude $U(\mathbf{K}, t)$ is in fact a random variable, as is the scattered field itself. The intensity of the scattered radiation is proportional to the mean-squared scattered field, while the spectrum of the radiation is determined by the temporal behavior of $U(\mathbf{K}, t)$. The information required to calculate the scattered power and the spectrum is contained in the correlation function of the scattered field $R_E(\tau)$ which is defined by [6]

$$\begin{aligned} R_E(\tau) &\equiv \langle \mathbf{E}_{\text{scatt}}(\mathbf{R}, t + \tau) \cdot \mathbf{E}_{\text{scatt}}^*(\mathbf{R}, t) \rangle \\ &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \mathbf{E}_{\text{scatt}}(\mathbf{R}, t + \tau) \cdot \mathbf{E}_{\text{scatt}}^*(\mathbf{R}, t) dt. \end{aligned} \quad (29)$$

The power $dP(\mathbf{K}, \mathbf{R})$ in all frequencies, which is scattered into the solid angle $d\Theta$ around θ is proportional to $R_E(0)$, i.e.,

$$\begin{aligned} dP_{\text{scatt}}(\mathbf{K}, \mathbf{R}) &= \frac{c}{4\pi} \langle |\mathbf{E}_{\text{scatt}}(\mathbf{R}, t)|^2 \rangle R^2 d\Theta \\ &= \frac{c}{4\pi} R^2 d\Theta R_E(0). \end{aligned} \quad (30)$$

The spectral density in the scattered radiation $S(\mathbf{K}, \omega)$ is given by

$$S(\mathbf{K}, \omega) \equiv \frac{1}{2\pi} \frac{\int_{-\infty}^{\infty} R_E(\tau) e^{i\omega\tau} d\tau}{R_E(0)}. \quad (31)$$

This is normalized in such a way that

$$\int_{-\infty}^{\infty} S(\mathbf{K}, \omega) d\omega = 1 \quad (32)$$

so that the power scattered into $d\Theta$ which lies between ω and $\omega + d\omega$ in frequency is given by

$$dP(\mathbf{K}, \omega) d\omega = \frac{c}{4\pi} R^2 d\Theta R_E(0) S(\mathbf{K}, \omega) d\omega. \quad (33)$$

To calculate the correlation function for the scattered field it is necessary to find the correlation function R_U

for the fluctuations in the Fourier amplitudes

$$R_U(\tau) \equiv \langle U(\mathbf{K}, t + \tau) \cdot U^*(\mathbf{K}, t) \rangle. \quad (34)$$

From our physical description of the temporal behavior of the thermal sound waves, it follows that the dependence of R_U on τ is simply

$$R_U(\tau) = \langle |U(\mathbf{K}, t)|^2 \rangle e^{\pm i\Omega(\mathbf{K})\tau} e^{-\Gamma(\mathbf{K})\tau} \quad (35)$$

where $\Omega(\mathbf{K})$ is the sound wave frequency and $\Gamma(\mathbf{K})$ is the decay rate of the wave with wave vector $\mathbf{K}$.

The only remaining unknown quantity is the mean-squared Fourier amplitude of the thermally excited sound waves. This can be related to the temperature as follows. The total vibrational energy of the liquid is equal, by the equipartition theorem for harmonic oscillators, to twice the vibrational kinetic energy.

$$\langle E \rangle = 2 \left\langle \int_V d\mathbf{r} \left| \frac{\rho_0}{2} \dot{U}(\mathbf{r}, t) \right|^2 \right\rangle. \quad (36)$$

On substituting (26) for $U(\mathbf{r}, t)$ into (36), we find, keeping in mind that for a single q there are two sound waves moving in opposite directions, that

$$\langle E \rangle = 2 \int |d\mathbf{q}| \rho_0 \Omega^2(q) \langle |U(\mathbf{q}, t)|^2 \rangle. \quad (37)$$

We may observe that (37) shows that the Fourier amplitudes $U(\mathbf{q}, t)$ are the normal coordinates for the sound vibrations. Denoting by $\langle E(q) \rangle$ the thermal average energy for each normal mode, we observe that the total energy is simply the sum over all normal mode energies. Writing this sum as an integral over $|d\mathbf{q}|$ with the appropriate density of states we have

$$\langle E \rangle = \int \frac{V}{(2\pi)^3} |d\mathbf{q}| \langle E(q) \rangle. \quad (38)$$

Comparison of (37) and (38) using $\langle E(q) \rangle = kT$ gives

$$\langle |U(\mathbf{K}, t)|^2 \rangle = \frac{V}{(2\pi)^3} \frac{kT}{2\rho_0 \Omega^2(\mathbf{K})}. \quad (39)$$

We now have all the information needed to find $R_E(\tau)$. Using (29), (16), (28), (35), and (39) we find

$$\begin{aligned} R_E(\tau) &= E_0^2 \left(\frac{\omega_0}{c} \right)^4 \frac{\sin^2 \phi}{(4\pi R)^2} \\ &\times \left[\left(\frac{\partial \epsilon}{\partial \rho} \right)_T + \left(\frac{\partial \epsilon}{\partial T} \right)_\rho \left(\frac{a}{\beta_T} \right) \frac{T}{\rho_0 c_V^*} \right]^2 \frac{\rho_0^2 kT}{2B_s} V \\ &\times \exp[-i(\omega_0 \mp \Omega(\mathbf{K}))\tau] \exp[-\Gamma(\mathbf{K})\tau] \end{aligned} \quad (40)$$

where B_s is the adiabatic bulk modulus $V(\partial P/\partial V)_s = \rho_0(\Omega/K)^2$. The $\pm$ designation denotes a sum over both the advancing and receding sound waves with wave vector $\mathbf{K}$. From this result we can now calculate the total power scattered in all directions. The incident power is P_0 , where $P_0 = A(cE_0^2/4\pi)$. A is the cross section of the incident beam. The illuminated volume V is equal to A times L , the length of the illuminated region.

From (30) and (40) it follows that the total power scattered in all directions is related to the incident power by

$$\frac{(P_{\text{scatt}})_{\text{Total}}}{P_0} = \left\{ \frac{\pi^2}{2} \left(\frac{1}{\lambda_0} \right)^4 \rho_0^2 \left[\left(\frac{\partial \epsilon}{\partial \rho} \right)_T + \left(\frac{\partial \epsilon}{\partial T} \right)_\rho \left(\frac{a}{\beta_T} \right) \frac{T}{\rho_0 c_V^*} \right]^2 \frac{kT}{B_s} \right\} \frac{8\pi L}{3}. \quad (41)$$

The quantity in braces is called the scattering coefficient [3]. If the incident light is 6328 Å, this coefficient ranges from $\sim 0.8 \times 10^{-6} \text{ cm}^{-1}$ for water, a very weak scatterer, to $\sim 40 \times 10^{-6} \text{ cm}^{-1}$ for CS_2 , a very strong scatterer [3].

The spectrum $S(\mathbf{K}, \omega)$ of the scattered light is given in accordance with (40) and (31) by

$$\begin{aligned} S(\mathbf{K}, \omega) &= \frac{1}{\pi} \left\{ \frac{\Gamma^2(\mathbf{K})}{[\omega - (\omega_0 + \Omega(\mathbf{K}))]^2 + \Gamma^2(\mathbf{K})} \right. \\ &\quad \left. + \frac{\Gamma^2(\mathbf{K})}{[\omega - (\omega_0 - \Omega(\mathbf{K}))]^2 + \Gamma^2(\mathbf{K})} \right\} \end{aligned} \quad (42)$$

where we have included the sum over the $\pm$ terms in (40). The spectrum of the scattered light thus consists of a pair of doublets, whose shape is Lorentzian if the correlation function for the sound-wave amplitude dies off exponentially like $\exp[-\Gamma(\mathbf{K})\tau]$. The width of the spectral lines gives the sound-wave lifetime ($1/\Gamma(\mathbf{K})$) and the splitting of the doublets around the unshifted frequency is

$$\omega' - \omega = \pm \Omega(\mathbf{K}). \quad (43)$$

This splitting gives the frequency of the sound wave having a wave vector equal to the scattering vector $\mathbf{K}$.

EXPERIMENTAL METHODS AND RESULTS

The system we employ to study Brillouin scattering from liquids is shown in the block diagram in Fig. 3. The constituents are 1) the laser light source, 2) the scattering cell and conical light gathering optics, 3) a Fabry-Perot spectrometer, and 4) a phototube detector and recorder.

The source of light is a short single-mode helium-neon laser with a power output of $\sim 1 \text{ mW}$. This laser was specially constructed to minimize the effects of temperature changes on the laser frequency. After thermal equilibrium is established the frequency excursion of this free running laser is less than $\sim 20 \text{ Mc/s}$ over a 15-minute period. It should be noted that with the aid of feedback devices the laser frequency can be stabilized much better than this. Frequency stabilized lasers are available commercially. While their frequency stability is much better ($\sim \pm 5 \text{ Mc/s}$) their power output is only about 0.2 mW .

The laser beam is focused into a narrow line, using a long focal length lens, and directed into the scattering cell containing the liquid under study. The liquids are carefully purified by distillation or by filtering with

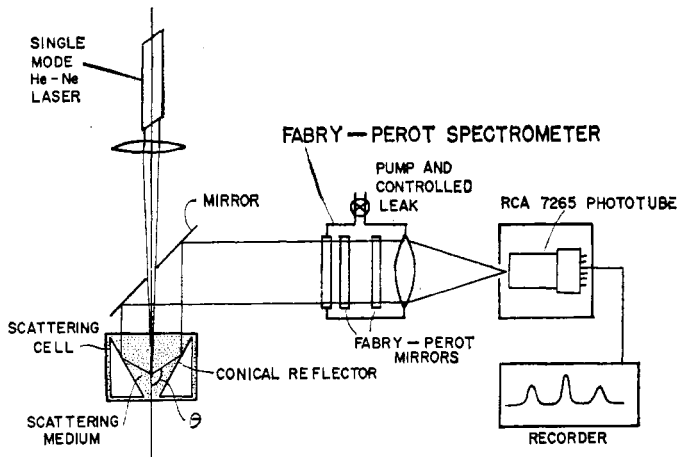


Fig. 3. Block diagram of the Fabry-Perot spectrometer.

Millipore filters to remove foreign suspended matter. This precaution is very valuable in minimizing the Tyndall scattering of light. To collect the light scattering an angle θ away from the forward direction we use a conical reflector inside the scattering cell. If ϕ is the full apex angle of the conical reflector, then the light scattered an angle $\theta = 180 - \phi$ will be collected by the cone and sent out parallel to the axis of the reflector. Of course, if the direction of the beam is reversed from that shown in Fig. 3, one collects light scattered through an angle ϕ . Thus a single reflector enables a study of scattering for two angles. This conical light-collection system has two advantages. First it collects the light over the full length of the illuminated region, and second, it collects light scattered a full 2π radians in azimuth. The conical geometry can, of course, be used with conical lenses in place of the conical reflectors discussed previously.

The light collected by the conical system and sent out parallel to the cone axis is analyzed into its spectral components by reflecting it into a pressure scanned Fabry-Perot spectrometer [7]. When the incident light is normal to the Fabry-Perot plates, as is the case in our system, the spectrometer acts like a narrow-band transmission filter whose passing frequency is swept by changing the gas density between the reflecting mirrors. To be more exact, the Fabry-Perot transmits not at one frequency, but at a host of frequencies. The transmitting windows are arranged like a picket fence with a frequency spacing f between windows, given by

$$f = (c/2nd) \quad (44)$$

where f is conventionally denoted as the free spectral range, d is the spacing between the reflecting plates, n is the index of refraction of the gas between the plates, and c is the speed of light. The ratio of the free spectral range to the width Δf of the transmission window is called the finesse F .

$$F = f/\Delta f. \quad (45)$$

F is determined by the flatness, the reflectivity, and the alignment of the mirrors. Practically, one can construct Fabry-Perot spectrometers with $F \lesssim 50$. Thus as an example for $d \sim 3$ cm the free spectral range is ~ 5 Gc/s and the width of the transmission window Δf is ~ 100 Mc/s. The spectrum of the incident radiation is analyzed by sweeping the transmission windows across the frequencies in the incident light. The transmission pattern as a whole is translated in frequency by permitting nitrogen gas to leak into the initially evacuated space between the mirrors. The pressure of the gas is arranged to increase linearly with the time, and the frequency of each window moves at a rate $d\nu/dt$ determined by

$$(d\nu/dt) = (\nu/n)(dn/dt). \quad (46)$$

Between zero and one atmosphere $\Delta n/n \sim 3 \times 10^{-4}$, so that the total excursion in ν is $\Delta\nu \sim 150$ Gc/s for the 6328 Å laser light. For the 3-cm Fabry-Perot example given previously each window moves thirty spectral ranges as the pressure sweeps from zero to one atmosphere. If light of frequency ν_0 and spectral width $\Delta\nu_0$ is incident on this Fabry-Perot it would pass through the system thirty times as the pressure sweeps from zero to one atmosphere. The passage of light through the spectrometer is detected by focusing the emergent light so that it passes through a small angle-defining aperture and falls on the surface of a phototube. The size of the admitting aperture is important in delineating the range of scattering angles $\Delta\theta$ that the system accepts.

The recorder traces out the transmission of light as the picket fence sweeps across the spectral distribution of the incident light. The recorder tracing can be calibrated easily in terms of frequency because the transmission pattern is periodic, with the periodicity equal to the free spectral range f . Thus, from the recorder tracing we can obtain the splitting of the Brillouin doublets, and, if Δf is less than or comparable to the natural width $\Delta\nu_0$, it also gives the spectral widths of the scattered light.

To demonstrate the use of this system we present in Fig. 4 a recorder tracing of the output of the Fabry-Perot spectrometer for light scattered by water through an angle of 172.1 degrees. Water is in fact one of the weakest liquid light scatterers. Another feature of the light scattering from pure water is the absence of an unshifted or central component in the scattered spectrum. The central component in the spectrum results from scattering off *nonpropagating* density fluctuations [8], [9]. We recently presented [10] an account of the theory of the central component and the first measurements of the width of the central component for a pure liquid near its critical point. Debye [11] gave the theory for the width of the central component in binary liquid mixtures, and Alpert, Yeh, and Lipworth [12] gave the first measurements of the width of this central component in a liquid mixture near the critical mixing temperature.

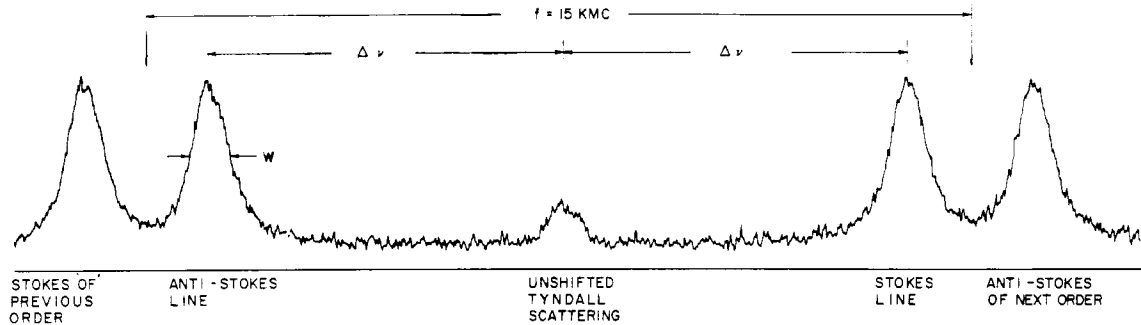


Fig. 4. The output of the Fabry-Perot spectrometer for light scattered by very pure water through 172.1 degrees at a temperature of 27.5° C. The full width at half height is a sum of the width due to natural decay of 260 Mc/s and an instrumental width of 470 Mc/s. The incident laser power was 0.14 milliwatt.

In Table I we give preliminary results of measurements of the velocity, the frequency, and the decay rate for ~6 Gc/s thermally excited sound waves in toluene and water.

TABLE I

THE VELOCITY AND DECAY RATE FOR HYPERSONIC SOUND WAVES IN WATER AND TOLUENE

Fluid	Temperature in °C	Scattering Angle in Degrees	Sound Wave Frequency in Mc/s	Sound Velocity in m/s	Decay Rate $\Gamma/2\pi$ (half width at half height) in Mc/s
Water	27.7	172.1°	6311 ± 6	1504 ± 1.4	130 ± 20
Toluene	22.2	172.9°	6485 ± 5	1379 ± 1.1	395 ± 15

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